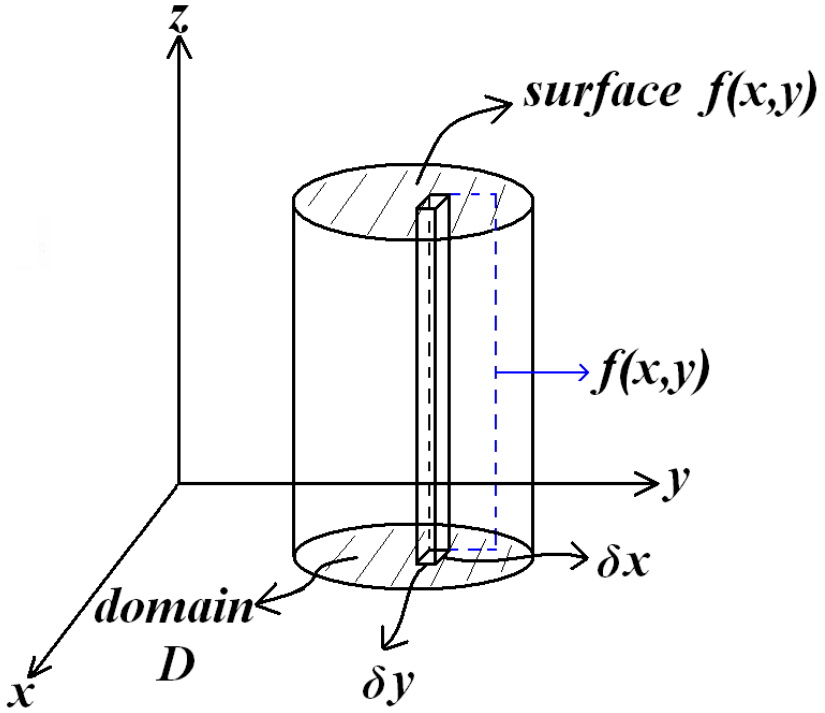
4. Double and Triple Integrals

4.18. Double Integrals

If f is a continuous function over a domain $D \subseteq \mathbb{R}^2$, we can evaluate the double integral

$$\iint_D f(x,y)\,dA = \iint_D f(x,y)\,dx\,dy.$$



$$\begin{aligned}\text{Volume of thin rod} &= \text{(Area base) (height)}\\ &= \delta A\, f(x,y)\\ &= \delta x\, \delta y\, f(x,y)\end{aligned}$$

Suppose we divide the region D into n small sections, with

- (x_i, y_i) being a point inside the ith section, and
- δA_i being the area of the ith section.

Then as $n \to \infty$, the sections get smaller and smaller and the Riemann sum becomes the integral

$$\begin{aligned}\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i, y_i)\delta A_i &= \iint_D f(x,y)\, dA\\ &= \iint_D f(x,y)\, dx\, dy\end{aligned}$$

Physical Interpretation

The double integral $\iint_D f(x, y)\, dA$ is the volume under the surface $z = f(x, y)$ that lies above the domain D in the xy-plane (assuming that $f(x, y) \geq 0$ over D).

Note

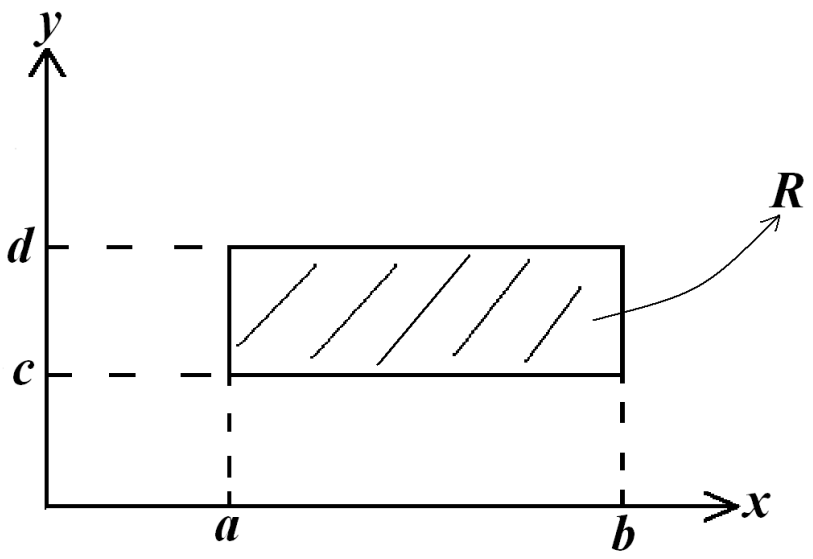
If $f(x, y) = 1$ then

$$\iint_D dA = \iint_D dx\, dy$$

is the total area of the domain D.

Double Integrals over Rectangular Domains

1. $R = [a, b] \times [c, d]$ is a rectangular domain defined by $a \leq x \leq b,\ c \leq y \leq d$.



2. $$\int_c^d \int_a^b f(x, y)\, dx\, dy \;=\; \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy$$
means integrate with respect to x first and then integrate with respect to y.

Fubini's Theorem

Let f be a continuous function over the domain $R = [a, b] \times [c, d]$. Then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y)\, dx\, dy = \int_a^b \int_c^d f(x, y)\, dy\, dx.$$

So the order of integration is NOT important.

Example 18.1

Evaluate $\iint\limits_R (x^2 + y^2)\, dx\, dy$ if $R = [-1, 1] \times [0, 1]$.

Example 18.2

Evaluate

$$\iint_R x\cos(xy)\,dA, \quad R = [0, \tfrac{\pi}{2}] \times [0, 1].$$

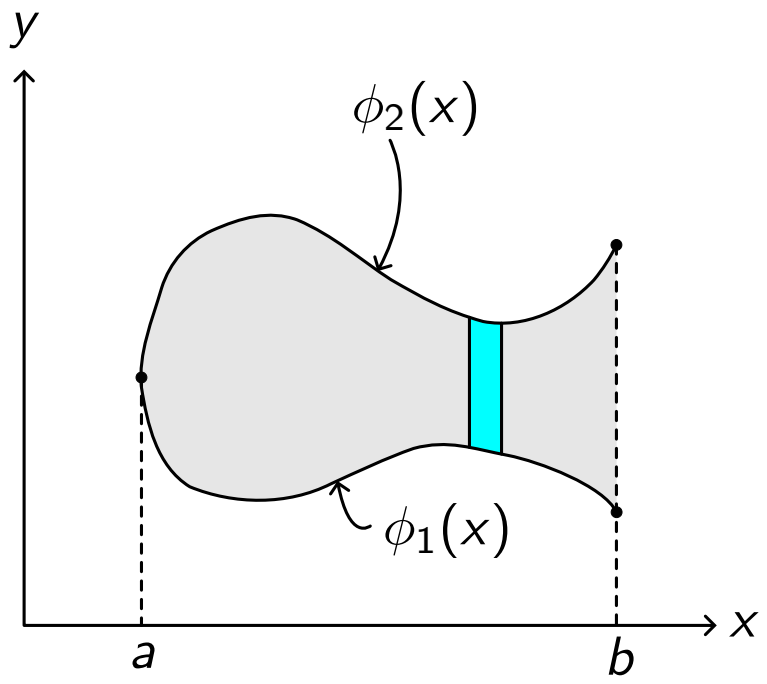
Double Integrals over General Domains

The domain determines the terminals and order of integration.

Type I region (vertically simple):

$$D = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, \phi_1(x) \leq y \leq \phi_2(x)\},$$

ϕ_1, ϕ_2 continuous on $[a, b]$.



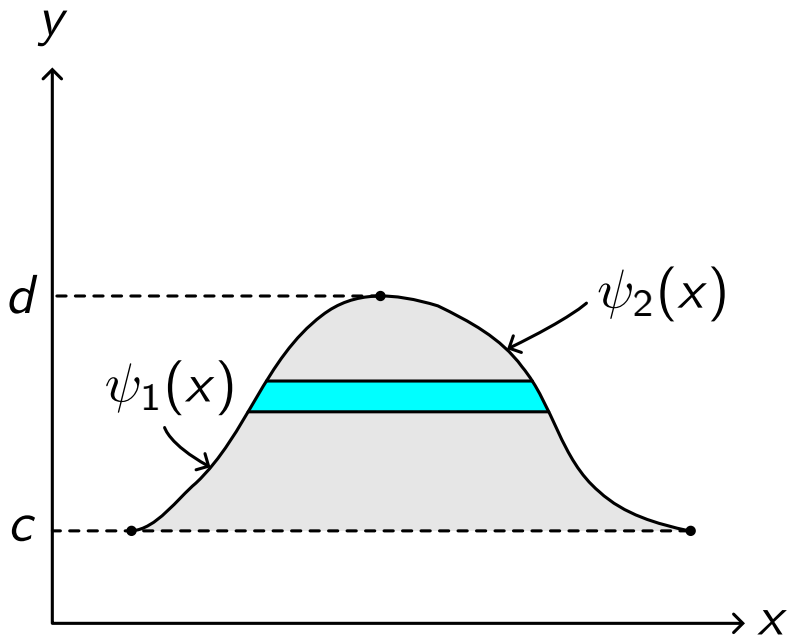
Integrate with respect to y first

$$\iint_D f(x, y) dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx$$

Type II region (horizontally simple):

$$D = \{(x, y) \in \mathbb{R}^2 : c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\},$$

ψ_1, ψ_2 continuous on $[c, d]$.



Integrate with respect to x first

$$\iint_D f(x, y) dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy$$

Example 18.3

Evaluate $\displaystyle\iint_D (x^3 y + \cos x)\, dA$

where D is the triangle with vertices at $(0,0), (\pi, 0)$ and (π, π).

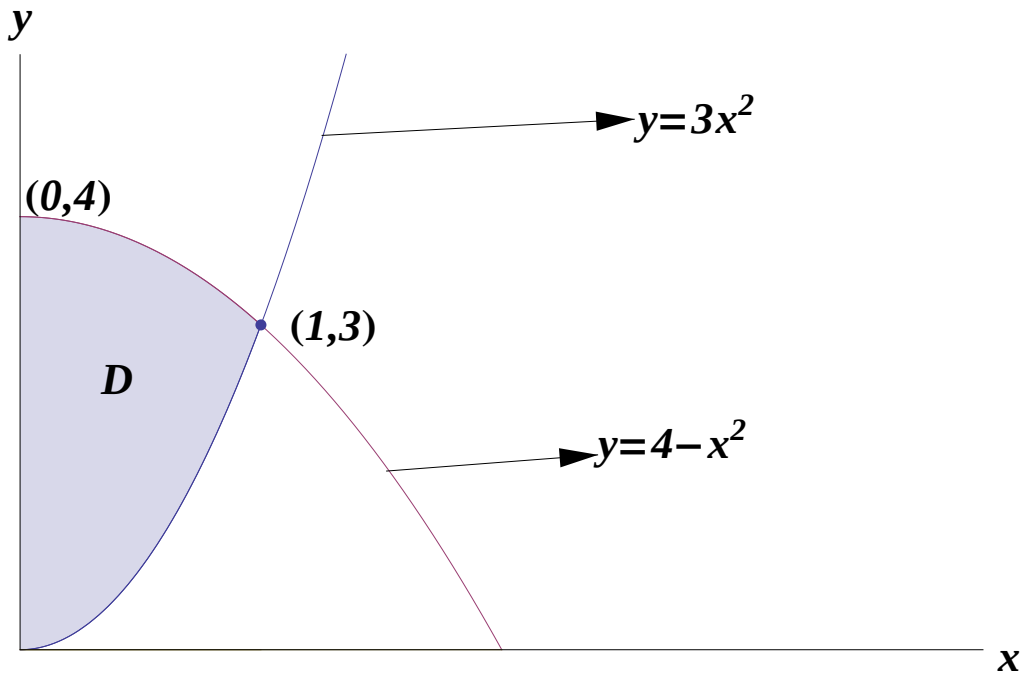
Note

The order of integration is not important if

1. the domain can be divided into horizontal and vertical strips AND
2. f is continuous in the domain.

Example 18.4

Find the area enclosed by $y = 3x^2$, $y = 4 - x^2$ and the y-axis for $x \geq 0$.



Note that this is also possible as a Type II region, but requires more work (has to be split into two parts).

Example 18.5

Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$.

Example 18.6

Evaluate $\int_0^2 \int_{x^3}^8 \cos\left(y^{\frac{4}{3}}\right)\, dy\, dx$.

Properties of double integrals

1. Linearity:

$$\iint_D (cf(x,y) + dg(x,y))\, dA = c \iint_D f(x,y)\, dA + d \iint_D g(x,y)\, dA$$

   where $c, d \in \mathbb{R}$ are constants.

2. Dominance rule: if $f(x,y) \geq g(x,y)$ for all $x, y \in D$, then

$$\iint_D f(x,y)\, dA \geq \iint_D g(x,y)\, dA.$$

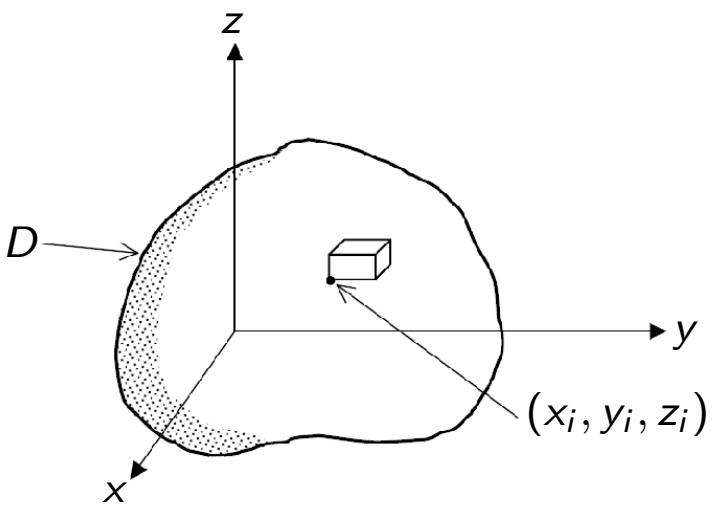
3. Subdivision rule: Suppose $D = D_1 \cup D_2$, where D_1 and D_2 do not overlap except perhaps on their boundaries. Then

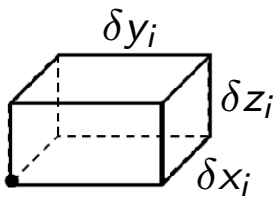
$$\iint_D f(x,y)\, dA = \iint_{D_1} f(x,y)\, dA + \iint_{D_2} f(x,y)\, dA.$$

4.19. Triple Integrals

If f is a continuous function over a solid domain $D \subseteq \mathbb{R}^3$, we can evaluate the triple integral

$$\iiint\limits_D f(x, y, z)\, dV \;=\; \iiint\limits_D f(x, y, z)\, dx\, dy\, dz$$





Suppose we divide the region D into n small sections, with

- (x_i, y_i, z_i) being a point inside the ith section, and
- δV_i being the volume of the ith section.

Then as $n \to \infty$, the sections get smaller and smaller and the Riemann sum becomes the integral

$$\begin{aligned}\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i, y_i, z_i)\ \delta V_i &= \iiint_D f(x, y, z)\ dV \\ &= \iiint_D f(x, y, z)\ dz\ dy\ dx\end{aligned}$$

Physical Interpretation

The physical meaning of $\iiint\limits_D f(x, y, z)\, dV$ depends on f.

Example 1

If $f(x, y, z)$ is the mass per unit volume (i.e. the density) of D, then

$$\iiint\limits_D f(x, y, z)\, dV$$

is the total mass of D.

Example 2

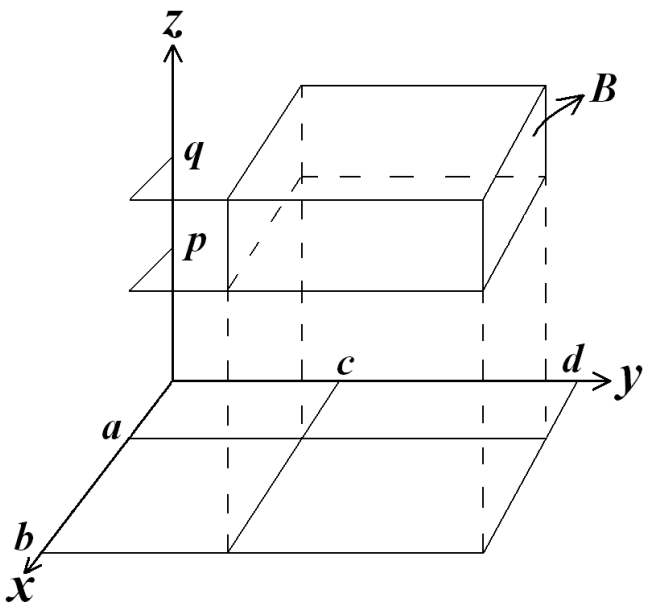
If $f(x, y, z) = 1$ then

$$\iiint_D dV = \iiint_D dx\, dy\, dz$$

is the volume of D.

Triple Integrals over Rectangular Box Domains

The rectangular box domain B



can be specified as

$$B = [a, b] \times [c, d] \times [p, q]$$
$$\text{or } a \leq x \leq b, \quad c \leq y \leq d, \quad p \leq z \leq q.$$

Fubini's Theorem

Let f be a continuous function over $B = [a, b] \times [c, d] \times [p, q]$, then

$$\begin{aligned}
\iiint_B f(x, y, z)\, dV &= \int_p^q \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz \\
&= \int_p^q \int_a^b \int_c^d f(x, y, z)\, dy\, dx\, dz \\
&= \int_a^b \int_c^d \int_p^q f(x, y, z)\, dz\, dy\, dx
\end{aligned}$$

and so on ($3! = 6$ possible combinations)

Example 19.1

Consider the solid box

$$B = [0,1] \times \left[-\frac{1}{2}, 0\right] \times \left[0, \frac{1}{3}\right]$$

with mass per unit volume

$$\mu(x,y,z) = x + 2y + 3z \ \ \mathrm{g/cm^3}.$$

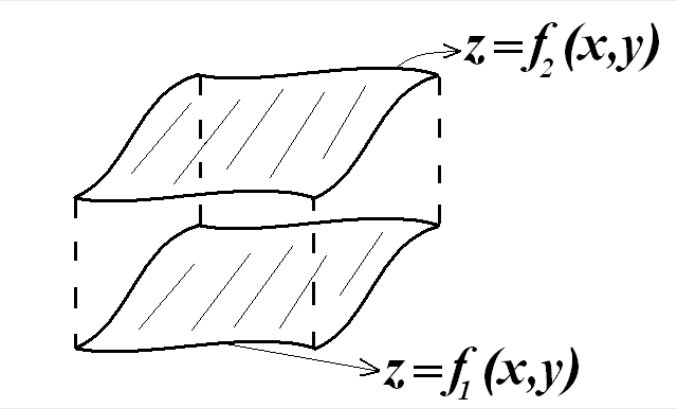
Determine the total mass of B.

Triple integrals over more general regions

The domain D is an elementary region in $\mathbb{R}^3$ if one variable is bounded by functions the other two variables, the domains of these functions being described using horizontal or vertical strips.

Example 19.2 (Type I region: z-simple)

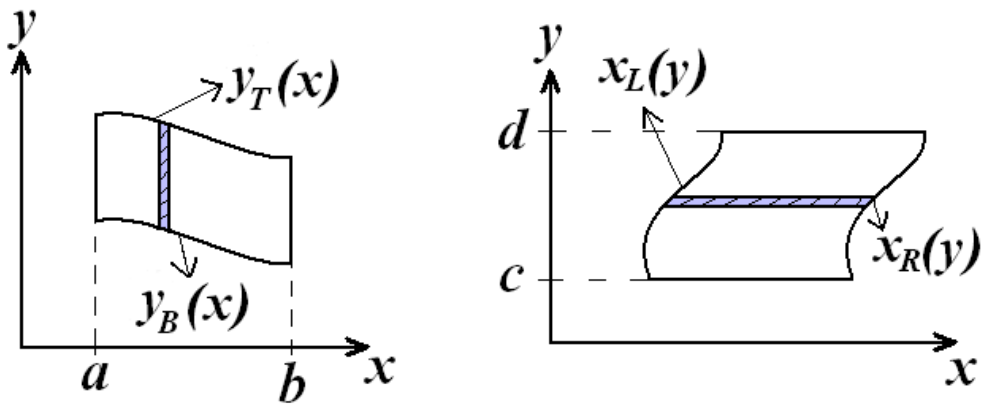
If D is a solid region bounded below by $z = f_1(x, y)$ and above by $z = f_2(x, y)$, that projects onto a region R in the xy-plane:



then

$$D = \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in R,\ f_1(x, y) \leq z \leq f_2(x, y)\}$$

The projection (or shadow) R of D onto the xy plane can then be described as before, using either vertical or horizontal strips.



Vertical strips **Horizontal strips**

So we have

$$R = \{(x, y) \in \mathbb{R}^2 : a \le x \le b,\ y_B(x) \le y \le y_T(x)\} \quad \text{(vertical strips)}$$

<u>OR</u>

$$R = \{(x, y) \in \mathbb{R}^2 : c \le y \le d,\ x_L(y) \le x \le x_R(y)\} \quad \text{(horizontal strips)}$$

Then integrate with respect to z first:

$$\iiint_D f(x, y, z)\, dV = \iint_R \left(\int_{f_1(x,y)}^{f_2(x,y)} f(x, y, z)\, dz \right) dA, \quad dA = dx\, dy$$

Example 19.3 (Type II region: x-simple)

$$D = \{(x, y, z) \in \mathbb{R}^3 : (y, z) \in R,\ x_1(y, z) \leq x \leq x_2(y, z)\}.$$

In this case D is bounded by two functions of y and z, and D projects onto a region R in the yz plane.

Integrate with respect to x first:

$$\iiint_D f(x, y, z)\, dV = \iint_R \left(\int_{x_1(y,z)}^{x_2(y,z)} f(x, y, z)\, dx \right) dA, \quad dA = dy\, dz$$

Example 19.4 (Type III region: y-simple)

$$D = \{(x, y, z) \in \mathbb{R}^3 : (x, z) \in R,\ y_1(x, z) \leq y \leq y_2(x, z)\}.$$

In this case D is bounded by two functions of x and z, and D projects onto a region R in the xz plane.

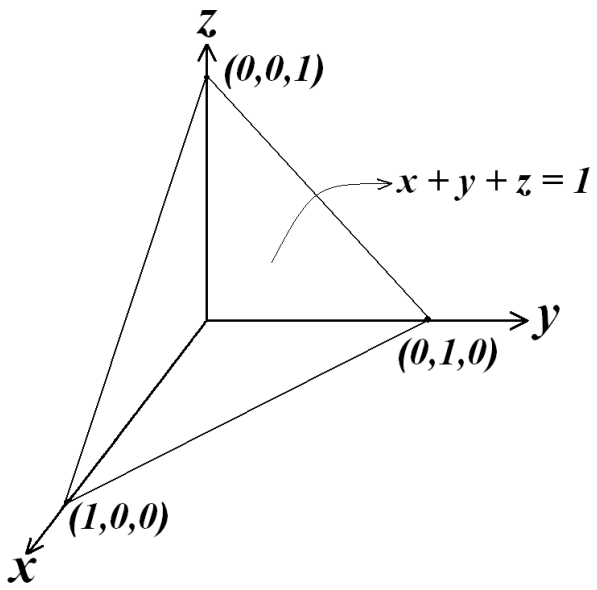
Integrate with respect to y first:

$$\iiint_D f(x, y, z)\, dV = \iint_R \left(\int_{y_1(x,z)}^{y_2(x,z)} f(x, y, z)\, dy \right) dA, \quad dA = dx\, dz$$

Example 19.5

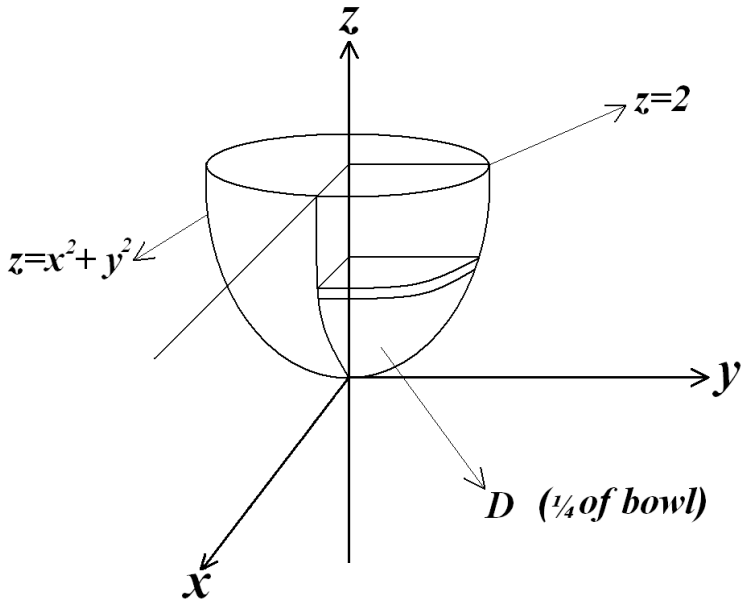
Evaluate $\iiint\limits_D xy\, dV$

where D is the solid tetrahedron with vertices at $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.



Example 19.6

Evaluate $\iiint\limits_D x\, dV$ where D is the solid region bounded by $z = 2$ and $z = x^2 + y^2$ in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$).



4.20. Change of variables in double integrals

Let D be an elementary region in $\mathbb{R}^2$ and suppose we wish to compute

$$\iint_D f(x,y)\,dA = \iint_D f(x,y)\,dx\,dy$$

Let us switch from (x,y) to different coordinates (u,v), by setting

$$x = x(u,v), \quad y = y(u,v).$$

(For example polar coordinates $x = u\cos v$, $y = u\sin v$.)

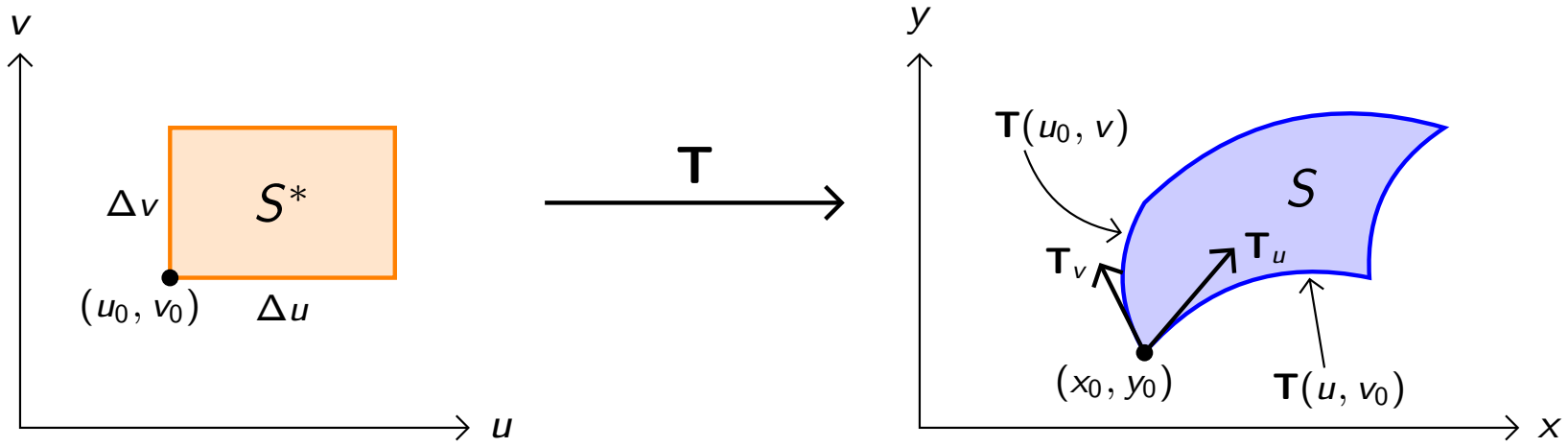
We can view this as a map $\mathbf{T} : D^* \to D$

$$\mathbf{T}(u,v) = (x(u,v), y(u,v))$$

where D^* is also a region in $\mathbb{R}^2$. Let us insist that $\mathbf{T}$ is C^1 and one-to-one.

What happens to the integral?

Consider a small rectangle S^* in D^*, with corner (u_0, v_0) and sides Δu and Δv. It maps to a small region S in D, with corner $\mathbf{T}(u_0, v_0) = (x(u_0, v_0), y(u_0, v_0)) = (x_0, y_0)$, and sides $\mathbf{T}(u, v_0)$ and $\mathbf{T}(u_0, v)$:



The tangent vector to the curve $\mathbf{T}(u, v_0)$ at (x_0, y_0) is

$$\mathbf{T}_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}\right)\Bigg|_{(u_0, v_0)}$$

and similarly the tangent vector to the curve $\mathbf{T}(u_0, v)$ at (x_0, y_0) is

$$\mathbf{T}_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}\right)\Bigg|_{(u_0, v_0)}$$

So we can approximate S by a parallelogram with sides $\Delta u \mathbf{T}_u$ and $\Delta v \mathbf{T}_v$.

The area of S is then approximately the area of the parallelogram, which is

$$|\Delta u \mathbf{T}_u \times \Delta v \mathbf{T}_v| = |\mathbf{T}_u \times \mathbf{T}_v| \Delta u \Delta v = \left| \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{bmatrix} \right| \Delta u \Delta v$$

$$= \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{bmatrix} \mathbf{k} \right| \Delta u \Delta v = \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{bmatrix} \right| \Delta u \Delta v$$

$$= \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \right| \Delta u \Delta v = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u \Delta v$$

We call

$$\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \frac{\partial(x, y)}{\partial(u, v)}$$

the Jacobian of the map $\mathbf{T}$.

Letting Δu and Δv approach 0, the corresponding Riemann sum becomes an integral and we can write

$$dA = \ dx\ dy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right|\ du\ dv$$

So finally

$$\begin{aligned}\iint\limits_D f(x,y)\ dA &= \iint\limits_D f(x,y)\ dx\ dy \\ &= \iint\limits_{D^*} f(x(u,v),y(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right|\ du\ dv\end{aligned}$$

where $\mathbf{T} : D^* \to D$ is C^1 and one-to-one.

Note that the Jacobian is the determinant of the matrix of partial derivatives, which may be positive or negative. However when we use it to change variables in an integral, we take the absolute value of this determinant.

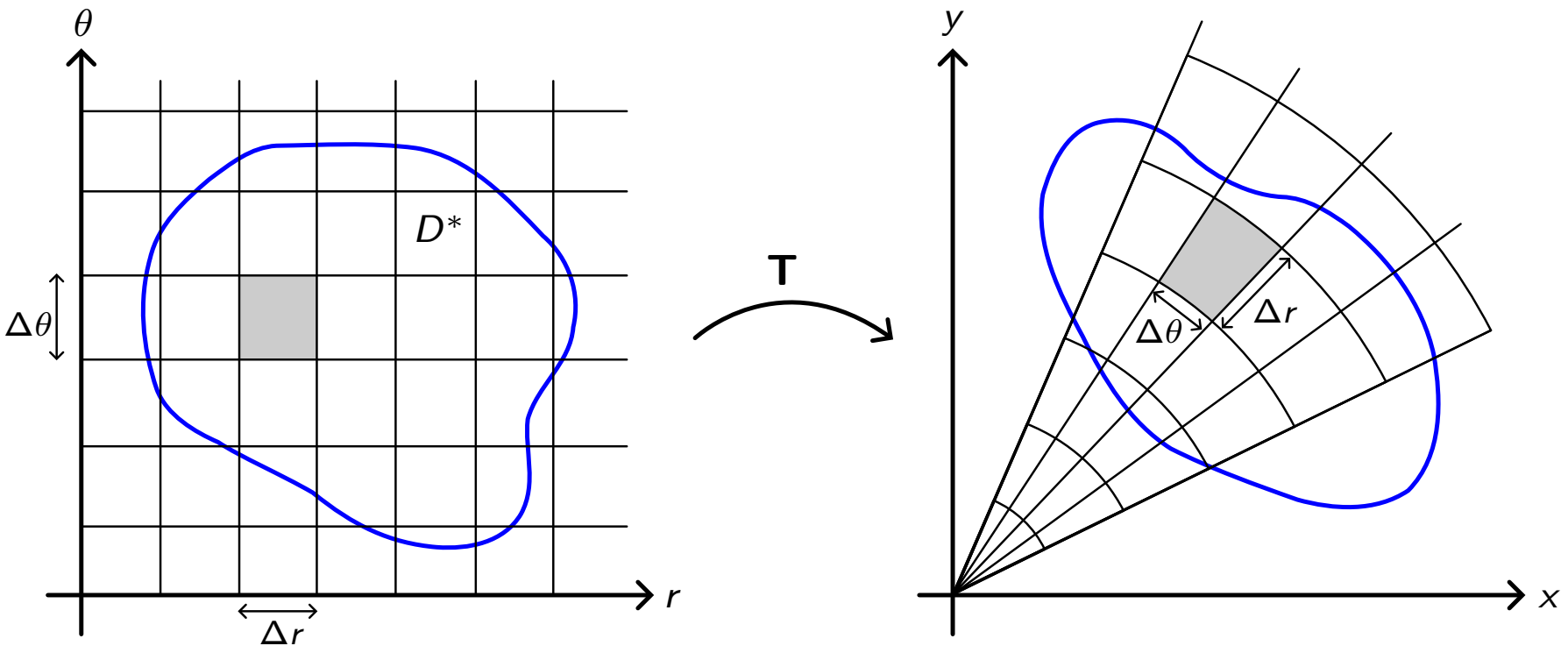
Note also that some references may use "Jacobian" to refer to the matrix of partial derivatives, rather than its determinant. Be careful!

Polar Coordinates (r, θ)

$$x = r\cos\theta, \; y = r\sin\theta \quad (r \geq 0)$$

$$\begin{aligned}
\text{Jacobian} &= \det \begin{bmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{bmatrix} \\
&= \det \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \\
&= r\cos^2\theta + r\sin^2\theta \\
&= r
\end{aligned}$$

$$\iint_D f(x, y)\, dx\, dy = \iint_{D^*} f(r\cos\theta, r\sin\theta) r\, dr\, d\theta$$


θ
D*
Δθ
Δr
r
T
y
Δθ
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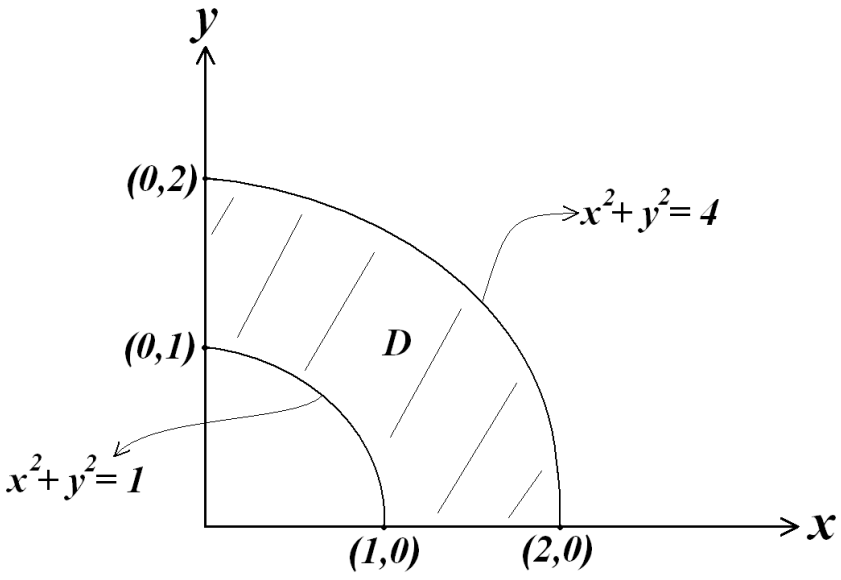
x

Example 20.1

Evaluate $\iint\limits_D \log(x^2 + y^2)\, dx\, dy$ where D is the region given below.

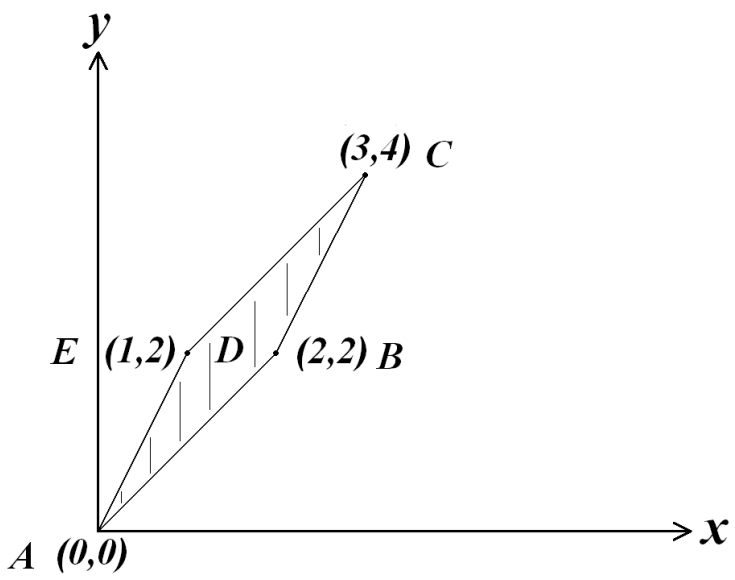


Example 20.2

Let D be the parallelogram with vertices $(0,0), (1,2)$, $(2,2)$, $(3,4)$.

Evaluate $\displaystyle\iint_D xy\, dx\, dy$ by making the change of variables

$x = u - v,\ y = 2u - v$.



Change of Variable Theorem for Triple Integrals

Let D, D^* be elementary regions in $\mathbb{R}^3$ and $T : D^* \to D$ be C^1. If T is one-to-one and $D = T(D^*)$

$$\iiint\limits_D f(x, y, z)\, dx\, dy\, dz$$

$$= \iiint\limits_{D^*} f\left[x(u, v, w), y(u, v, w), z(u, v, w)\right] \cdot \left|\frac{\partial(x, y, z)}{\partial(u, v, w)}\right| du\, dv\, dw$$

where the $|\text{Jacobian}|$ is

$$\left|\frac{\partial(x, y, z)}{\partial(u, v, w)}\right| = \left|\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}\right|.$$

Cylindrical Coordinates (ρ, ϕ, z)

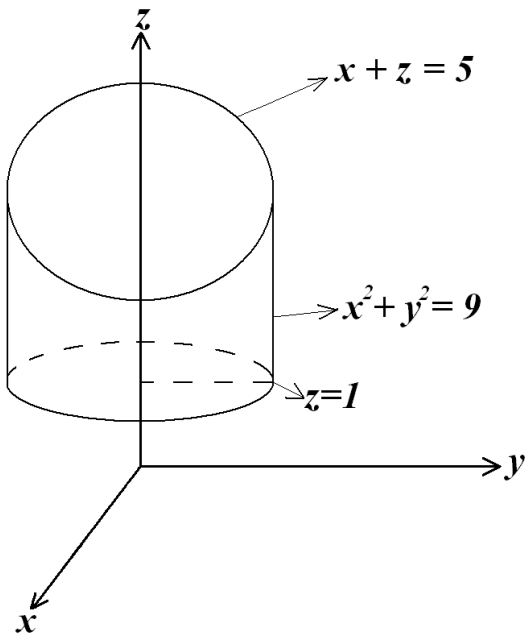
$$x = \rho\cos\phi, \; y = \rho\sin\phi, \; z = z \quad (\rho \geq 0)$$

$$\begin{aligned}
\text{Jacobian} &= \det \begin{bmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \phi} & \frac{\partial z}{\partial z} \end{bmatrix} \\
&= \det \begin{bmatrix} \cos\phi & -\rho\sin\phi & 0 \\ \sin\phi & \rho\cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \det \begin{bmatrix} \cos\phi & -\rho\sin\phi \\ \sin\phi & \rho\cos\phi \end{bmatrix} \\
&= \rho\cos^2\phi + \rho\sin^2\phi \\
&= \rho
\end{aligned}$$

$$\Rightarrow \iiint\limits_{D} f(x, y, z)\, dx\, dy\, dz = \iiint\limits_{D^*} f(\rho\cos\phi, \rho\sin\phi, z)\rho\, d\rho\, d\phi\, dz$$

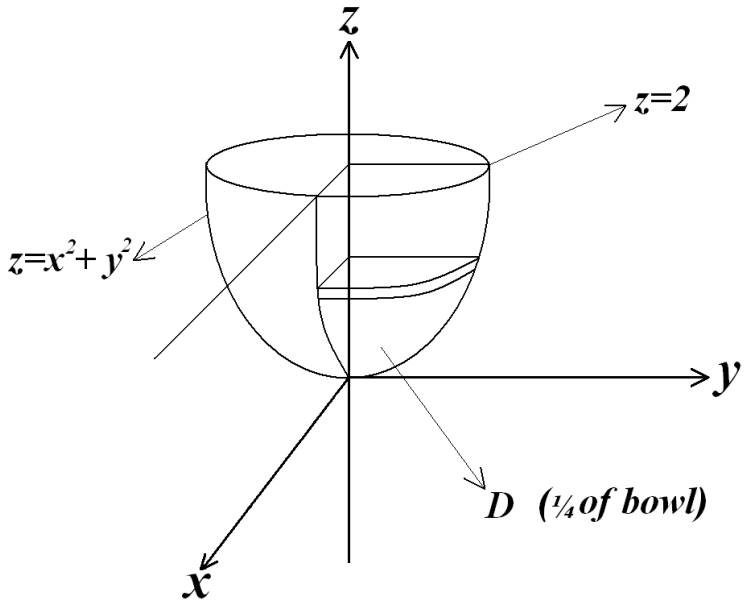
Example 20.3

Find the volume of the solid region enclosed by the cylinder $x^2 + y^2 = 9$ and the planes $z = 1,\ x + z = 5$.



Example 20.4

Redo Example (19.6) from triple integrals to find $\iiint\limits_D x\, dV$ for region below.



Spherical Coordinates (r, θ, ϕ)

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta \quad (r \geq 0)$$

$$\text{Jacobian} = \det\begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{bmatrix}$$

$$= \det\begin{bmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{bmatrix}$$

expand along third row:

$$\begin{aligned} &= \cos\theta\left[r^2\cos\theta\sin\theta\cos^2\phi + r^2\sin\theta\cos\theta\sin^2\phi\right] \\ &\quad + r\sin\theta\left[r\sin^2\theta\cos^2\phi + r\sin^2\theta\sin^2\phi\right] \\ &= r^2\sin\theta\cos^2\theta + r^2\sin^3\theta \\ &= r^2\sin\theta \qquad (\sin\theta \geq 0) \end{aligned}$$

$$\Rightarrow \iiint_D f(x,y,z)\,dx\,dy\,dz = \iiint_{D^*} F(r,\theta,\phi) r^2 \sin\theta\,dr\,d\theta\,d\phi$$

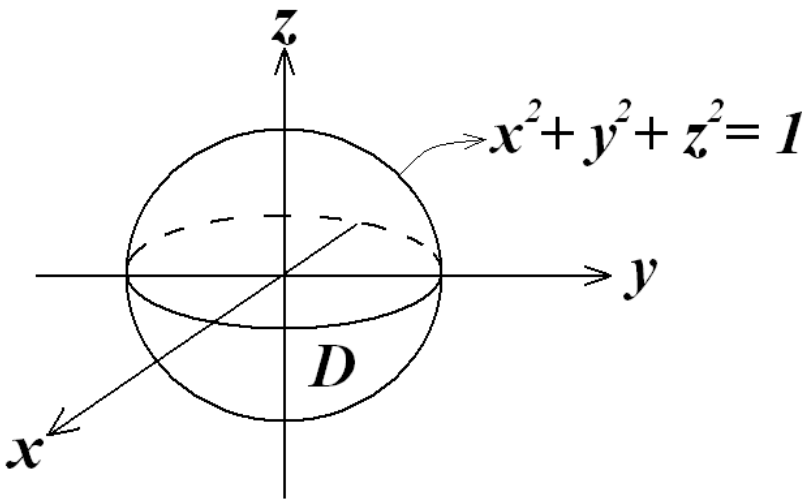
where

$$F(r,\theta,\phi) = f(r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$$

Example 20.5

Let D be the unit ball centred at $(0,0,0)$. Determine the mass of D if the mass per unit volume of D is

$$\mu(x,y,z) = \exp\left(\left[x^2+y^2+z^2\right]^{\frac{3}{2}}\right).$$



Note

If terminals are constant and the integrand factorises, i.e.

$$g(x, y, z) = g_1(x)g_2(y)g_3(z),$$

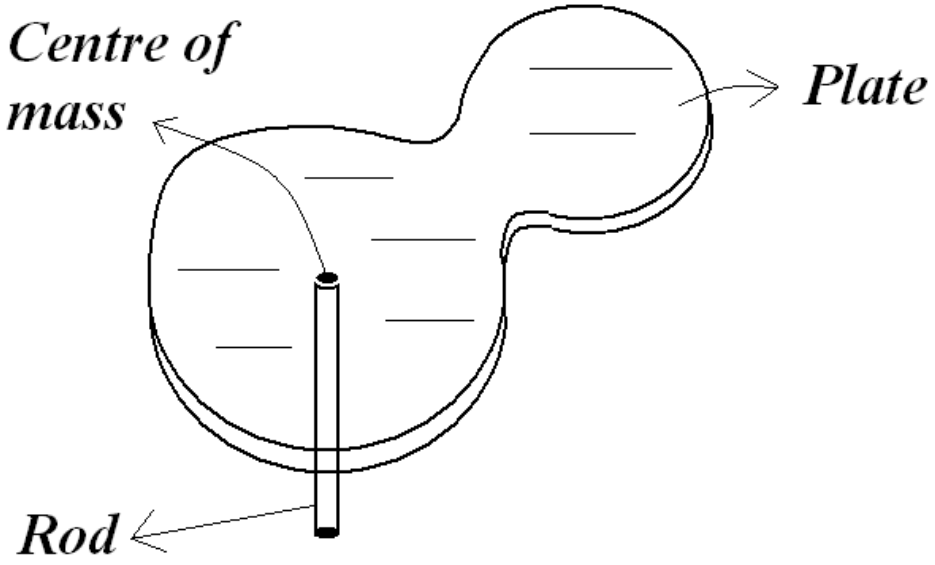
then

$$\int_a^b \int_c^d \int_e^f g(x, y, z)\, dx\, dy\, dz$$
$$= \left[\int_a^b g_3(z)\, dz\right] \left[\int_c^d g_2(y)\, dy\right] \left[\int_e^f g_1(x)\, dx\right].$$

4.21. Applications of Multiple Integrals

Centre of mass

The centre of mass of a rigid body is the point at which the body can be supported so it does not experience any unbalanced torques that cause it to rotate.



The plate balances when supported at its centre of mass.

- For a 2D plate with mass per unit area $\mu(x, y)$, the centre of mass is (x_{cm}, y_{cm}) where

$$x_{cm} = \frac{\iint_D x\,\mu(x,y)\,dx\,dy}{\text{mass of plate}}$$

$$y_{cm} = \frac{\iint_D y\,\mu(x,y)\,dx\,dy}{\text{mass of plate}}$$

where

$$\text{mass of plate} = \iint_D \mu(x,y)\,dx\,dy.$$

Note

If you have done any probability theory, compare the equations above to the expression for the expected value of a continuous random variable.

- Similarly for a 3D body with mass per unit volume $\mu(x, y, z)$, the centre of mass is (x_{cm}, y_{cm}, z_{cm}) where

$$x_{cm} = \frac{\iiint\limits_D x\,\mu(x, y, z)\,dx\,dy\,dz}{\text{mass of body}}$$

$$y_{cm} = \frac{\iiint\limits_D y\,\mu(x, y, z)\,dx\,dy\,dz}{\text{mass of body}}$$

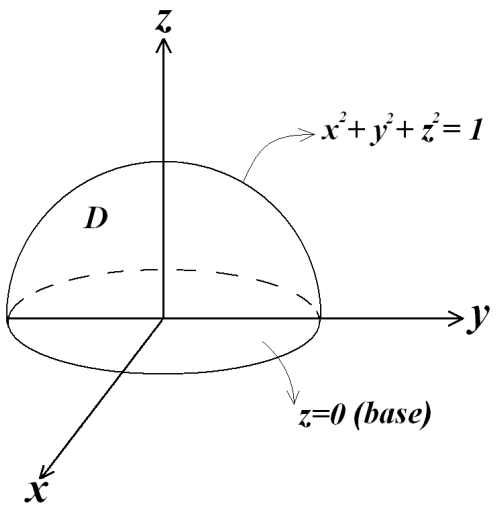
$$z_{cm} = \frac{\iiint\limits_D z\,\mu(x, y, z)\,dx\,dy\,dz}{\text{mass of body}}$$

where

$$\text{mass of body} = \iiint\limits_D \mu(x, y, z)\,dx\,dy\,dz.$$

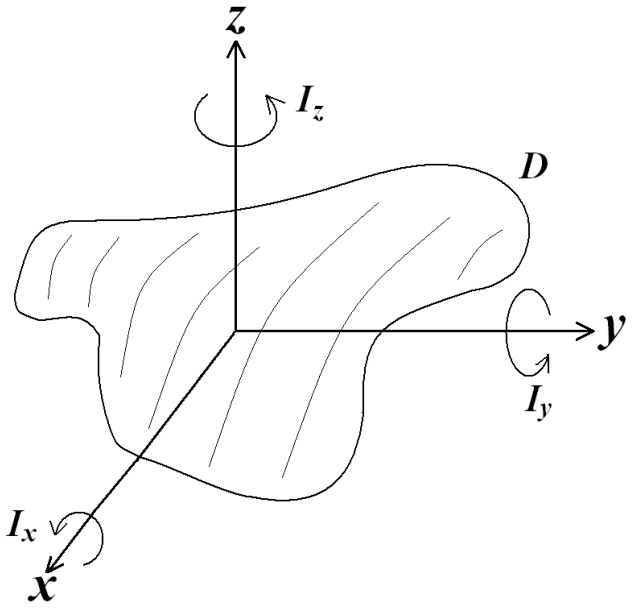
Example 21.1

Find the centre of mass of a solid hemisphere of radius 1 centred at $(0, 0, 0)$ for $z \geq 0$, if the mass per unit volume μ is constant.



Moments of Inertia

- Let I_n denote the moment of inertia of a solid body about the n-axis.
- I_n measures a body's response to spinning it about the n-axis



As I_n increases, it becomes harder to spin the body about the n-axis.

- If $\mu(x, y, z)$ is the mass per unit volume of a solid body, then

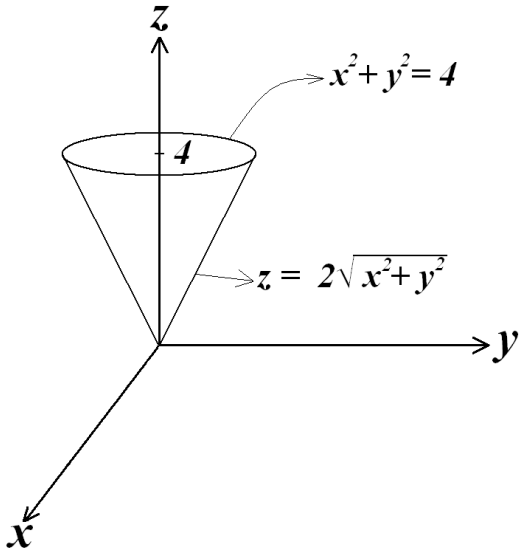
$$
\begin{aligned}
I_x &= \iiint_D (y^2 + z^2)\mu(x, y, z)\, dx\, dy\, dz \\
I_y &= \iiint_D (x^2 + z^2)\mu(x, y, z)\, dx\, dy\, dz \\
I_z &= \iiint_D (x^2 + y^2)\mu(x, y, z)\, dx\, dy\, dz
\end{aligned}
$$

Example 21.2

Suppose the density of the material inside the solid cone

$$z = 2\sqrt{x^2 + y^2},\ z \leq 4$$

varies as

$$\mu(x, y, z) = 5 - z.$$

Find the moment of inertia of the cone about the z-axis.